

Project Size Threshold

< \$5M → small
\$5M–\$25M → medium
> \$25M → large

Material Spec Examples

cost_items
material_type -> PVC

material_spec
c900
c905
sch40
sch80

Pipe Use Examples

force_main
gravity_sewer
potable_water
reclaimed_water
storm_drain

Delivery Method Examples

design_bid_build
design_build
cmar

Unit Code Examples

LF
SY
EA
TON
CY

Pressure Class Examples

DR18
DR25
Class 350

Item Category Examples

Pipe
Valve
Manhole
Pavement
Concrete

Agencies Examples

FDOT
JEA
Hillsborough County
City of Tampa

Project Type Examples

water
sewer
stormwater
roadway
site
general

Role Name Examples

admin
user

Permission Name Examples

view_cost_data
upload_projects
manage_users
purge_data
export_data

Database shapes

- Existing table
- Proposed table

bid_items	
bid_item_id	bigint (PK)
item_description	text
quantity	numeric(12,2)
unit_id	FK
unit_price	numeric(10,2)
project_id	FK
cost_item_id	FK
contractor_id	FK
created_at	timestamp
source_id	FK

projects	
project_id	PK
project_name	text
project_number	text
agency_id	FK
region_id	FK
project_type_id	FK
bid_date	date
project_manager	varchar(255)
engineer_of_record	varchar(255)
construction_cost	numeric(14,2)
delivery_method	text
project_size_id	FK

agencies	
agency_id	PK
agency_name	text
state	char(2)
city	text
region_id	FK

users	
user_id	integer (PK)
username	varchar(25)
email	varchar(255)
password_hash	text
is_active	bool
created_at	timestamp
last_login	timestamp

Entity	
Field	Type
Field	Type
Field	Type
Field	Type

project_types	
project_type_id	integer(PK)
type_name	varchar(25)

pipe_use	
pipe_use_id	integer(PK)
pipe_use_name	varchar(25)
description	text

contractors	
contractor_id	PK
contractor_name	text
state	char(2)
city	text
is_prequalified	bool

roles	
role_id	integer(PK)
role_name	varchar(25)
description	text

Entity	
Field	Type
Field	Type
Field	Type
Field	Type

cost_items	
cost_item_id	PK
pipe_use_id	FK
canonical_description	text
spec_section	varchar(15)
material_type	varchar(25)
material_spec	varchar(25)
pipe_diameter	integer
pressure_rating	varchar(15)
item_category	varchar(50)
installation_type	varchar(25)

units	
unit_id	integer(PK)
unit_code	char(5)
unit_description	text

regions	
region_id	integer(PK)
region_name	varchar(25)
state	char(2)
cost_index	numeric(5,2)

project_size	
project_size_id	integer(PK)
size_name	varchar(25)
description	text

permissions	
permissions_id	integer(PK)
permission_name	varchar(25)
description	text

user_roles	
user_id	PK
role_id	PK

audit_logs	
audit_id	bigint(PK)
user_id	FK
action	text
table_name	varchar(25)
record_id	FK
timestamp	timestamp
details_json	jsonb

data_sources	
data_source_id	integer(PK)
source_name	varchar(25)
source_type	varchar(15)
file_name	text
upload_date	timestamp
processed_by	varchar(255)

Add different endpoints by selecting a line and choosing from the dropdown in the top panel.



Customize class shapes by selecting them and editing them in the contextual panel that appears.